BASE CHANGE ISOMORPHISM FOR DERIVED PUSHFORWARD ALONG THE LISSE-ETALE (RESP. FLAT-FPPF) REFINEMENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author omits the verification of the assumptions of the cohomology-on-subcategory lemma.

Allowed standard prerequisites (editorial).

- Pushforward formula and cohomology on subcategories. Sheaves on Stacks, Lemma stacks-sheaves-lemma-pushforward-restriction and stacks-sheaves-lemma-cohomology-on-subcategory. Whole argument.
- delivered companion: higher direct image formula on sub-sites. delivered companion lemma-lisse-etale-functorial-pushforward. Second formula.

Source reference. Original lemma 07AT: sources/stacks-chow/tags.txt, source lines 9318–9318. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075G: sources/stacks-chow/tags.txt, source lines 9130–9130. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GR2: sources/stacks-chow/tags.txt, source lines 20882–20882. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07AK: sources/stacks-chow/tags.txt, source lines 9310–9310. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *With assumptions and notation as in Lemma 07AT the canonical (base change) map*

$$g^{-1}Rf_*\mathcal{F} \longrightarrow Rf'_*(g')^{-1}\mathcal{F}$$

is an isomorphism for any abelian sheaf $\mathcal{F}$ on $\mathcal{X}_{étale}$ (resp. $\mathcal{X}_{fppf}$).

Proof. Comparing the formula for $g^{-1}R^pf_*\mathcal{F}$ and $R^pf'_*(g')^{-1}\mathcal{F}$ given in Sheaves on Stacks, Lemma 075G and Lemma 0GR2 we see that it suffices to show

$$H^p((V \times_{y,\mathcal{Y}} \mathcal{X})',\ \mathrm{pr}^{-1}\mathcal{F}|_{(V\times_{y,\mathcal{Y}}\mathcal{X})'}) = H^p_\tau(V \times_{y,\mathcal{Y}} \mathcal{X},\ \mathrm{pr}^{-1}\mathcal{F})$$

where $\tau = étale$ (resp. $\tau = fppf$). Here y is an object of $\mathcal{Y}$ lying over a scheme V such that the morphism $y : V \to \mathcal{Y}$ is smooth (resp. flat). This equality follows from Sheaves on Stacks, Lemma 07AK. Although we omit the verification of the assumptions of the lemma, we note that the fact that $V \to \mathcal{Y}$ is smooth (resp. flat) is used to verify the second condition. □

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